6 Months Training TR-104 Full Stack Development

January, 2026

Day 5 – Template Control Flow and Conditional Rendering

1. Objective of the Day

The objective of the day was to understand Angular's modern template control flow and learn how to dynamically render UI elements using conditional statements and loops.

2. Tasks Performed

- Implemented conditional rendering using @if and @else
- Understood the difference between expressions and statements in Angular templates
- Practiced showing and hiding UI elements based on conditions
- Implemented looping using @for to display dynamic lists
- Rendered list data stored in signals
- Used track keyword for efficient list rendering
- Added and removed items dynamically from arrays
- Observed automatic UI updates when data changed

3. Tools / Technologies Used

- Angular Framework
- TypeScript
- HTML

4. Key Learnings

- Angular templates use expressions, not statements
- @if and @else control UI rendering dynamically

- @for is used to generate repeated UI elements
- UI automatically updates when underlying data changes
- Track improves performance in list rendering

5. Challenges Faced

- Confusion between TypeScript logic and template control flow
- Difficulty in understanding correct usage of @for with tracking

6. Solutions

- Practiced multiple examples differentiating logic and UI rendering
- Implemented sample lists to understand tracking behavior

7. Conclusion

- Successfully implemented conditional rendering and dynamic lists
- Gained clarity on Angular's declarative UI approach